

Executive Summary

Thyroid Nodule TI-RADS Classification Using Ultrasound Images

Thyroid nodules are extremely common, but most are benign. The clinical challenge is not finding nodules but figuring out which ones are worth worrying about. Radiologists use the TI-RADS scoring system to guide that decision, but ultrasound interpretation is subjective and varies between clinicians. A reliable classification model gives radiologists a consistent, data-backed second opinion and helps prioritize the cases that genuinely need follow-up.

What We Built

A deep learning pipeline that takes thyroid ultrasound images and patient metadata as inputs and classifies nodules into three clinically meaningful risk categories: benign, low-risk, and high-risk.

What We Found

Training on all seven original TI-RADS categories produced 60% accuracy. The model struggled because several mid-range categories share overlapping visual features in ultrasound images and have very few examples to learn from. Consolidating to three categories aligned the model with how clinical decisions are actually made. The relevant question for a radiologist is not whether a nodule is TI-RADS 4a versus 4b. It is whether the nodule warrants a biopsy. After consolidation, accuracy improved to 85%!

The most important result is high-risk recall. The model correctly identified 56 out of 57 truly high-risk nodules, a recall of 98%. Missing a high-risk nodule is the costliest error in thyroid screening. Getting 98 out of 100 right makes this model a credible candidate for clinical decision support.

We also tested DenseNet121, a pretrained model built on natural photographs. It did not adapt well to ultrasound images despite fine-tuning. The custom CNN trained from scratch on the actual dataset outperformed it, reinforcing that domain-specific training matters more than model complexity in medical imaging.

What This Means

A model with 98% high-risk recall and 85% overall accuracy can help radiologists work more consistently and flag the nodules that need attention faster. It does not replace clinical judgment. It supports it by reducing the chance that a dangerous nodule gets missed in a busy reading environment.

Limitations

The dataset contains 400 patients, which is small for deep learning. Low-risk nodules remain the hardest category to classify reliably, with 25% recall, reflecting genuine visual ambiguity that affects human readers as well. Validation on an independent external dataset is needed before any clinical deployment.

